

NEURAL TANGENT WOODBURY EXACT AUCTIONS WITH INEQUALITY-AVERSE SOCIAL WELFARE

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ABSTRACT

Machine learning classifiers deployed in economic decision-making implicitly assume uniform user valuations, leading to severe social welfare sub-optimality when individual utilities vary across demographic and socioeconomic groups. While accuracy auctions provide a mechanism to elicit private user valuations through truthful monetary bidding, existing mechanisms suffer from five core methodological bottlenecks: non-convex optimization trajectories breaking dominant-strategy incentive compatibility (DSIC), numerical pricing approximation errors, wealth bias favoring high earners, vulnerability to strategic feature manipulation, and linear payment blowups ($\Omega(m)$) in non-stable architectures. In this work, we present the **Neural Tangent Woodbury Exact Auction with Kolm-Pollak Welfare (NTW-KPA)**, a unified theoretical and algorithmic framework resolving all five limitations. NTW-KPA integrates Kolm-Pollak absolute inequality-averse utility mapping with ex-ante budget caps to eliminate wealth distortion and guarantee ex-post Individual Rationality ($p_i \leq B_i$). To support non-linear representations without non-convexity, inputs are projected into an infinite-width Neural Tangent Kernel (NTK) Hilbert space with a localized distance Gaussian RBF defense kernel, proving uniform RKHS classification stability $\beta = O(1/m)$ and establishing a constant $O(1)$ total population payment upper bound. Finally, using rank-1 Woodbury matrix identities, NTW-KPA derives an exact scalar rational path solver $\hat{y}_i(z) = \frac{A_i z + B_i}{C_i z + D_i}$ that computes exact zero-margin critical bids in a single scalar step with zero prediction discrepancy ($< 7.28 \times 10^{-15}$). Empirical evaluations on synthetic Gaussian and real-world ACSIncome benchmarks demonstrate a $17.2\times$ payment runtime speedup, flat $O(1)$ payment scaling matching linear SVMs, and a significant reduction in strategic feature gaming flip rates (0.377 vs 0.501). A low-rank primal reformulation further scales the exact solver to $n = 40,000$ users in 3.84 s and 332 MB (versus dense out-of-memory), while an anisotropic Mahalanobis defense kernel separates non-causal feature gaming from genuine outcome-improving effort by a $3.4\times$ wider margin. These guarantees hold across seven additional credit-scoring and multi-state census benchmarks and remain robust to noisy causal-cost estimates, while the closed-form payment speedup widens to $> 10^5\times$ in high-dimensional feature spaces.

1 INTRODUCTION

Machine learning classifiers are increasingly deployed to automate high-stakes economic decisions, ranging from loan approvals and employment screening to public resource distribution and health-care allocation. Standard empirical risk minimization (ERM) paradigms implicitly optimize unweighted classification accuracy, assuming that all individuals derive uniform utility from correct predictions Kleinberg & Raghavan (2018); Athey & Wager (2017). In heterogeneous real-world populations, however, user valuations for predictive accuracy vary significantly across demographic and socioeconomic groups Hu & Chen (2017); Heidari et al. (2019). When individual valuations are private and heterogeneous, unweighted accuracy maximization leads to severe social welfare sub-optimality, as decision boundaries fail to prioritize users with high stakes Sagawa et al. (2020);

Zhan et al. (2021). To align predictive models with population-level economic welfare, recent research has explored algorithmic mechanism design Akbarpour et al. (2023); Chen et al. (2023) and accuracy auctions Liu et al. (2024); Jin et al. (2022), wherein strategic agents report monetary bids reflecting their valuations to guide model training.

Despite their theoretical promise, existing accuracy auctions and welfare-maximizing classification mechanisms suffer from five core methodological bottlenecks:

1. **Non-Convex Monotonicity Violations:** Non-convex SGD trajectories break Myerson’s allocation monotonicity Zhou & Zhang (2024); Feng et al. (2018), destroying dominant-strategy incentive compatibility (DSIC).
2. **Pricing Approximation Errors:** Critical thresholds computed via binary-search grid re-training ($O(m \log(1/\epsilon))$) or local sensitivity approximations fail to locate exact zero-margin roots, introducing truthfulness-violating pricing errors.
3. **Wealth Bias and Unenforced Budgets:** Direct linear bid weighting ($w_i = b_i$) shifts boundaries toward wealthy agents, distorting democratic allocation and ignoring hard budget constraints (B_i).
4. **Strategic Feature Manipulation:** Users manipulate features ($x_i \rightarrow x'_i$) and bids (b_i) to reduce payments while keeping favorable predictions Dong et al. (2017), bypassing linear penalty defenses.
5. **Payment Blowups in Non-Stable Models:** Non-stable architectures (k -NN, standard nets) oscillate under label noise, causing total payments to scale linearly ($\Omega(m)$) rather than remaining bounded.

To resolve all five limitations simultaneously, we propose the **Neural Tangent Woodbury Exact Auction with Kolm-Pollak Welfare (NTW-KPA)**. NTW-KPA unifies social choice theory, infinite-width kernel representations, localized strategic distance kernels, and exact rank-1 matrix update identities into a provably truthful classification auction framework:

- **Kolm-Pollak Inequality-Averse Welfare Mapping:** Raw bids b_i are transformed via the Kolm-Pollak utility $U_\gamma(b) = (1 - e^{-\gamma b})/\gamma$ ($\gamma > 0$); setting training weights $w_i = \min(U_\gamma(b_i), U_\gamma(B_i))$ ex-ante eliminates wealth bias and guarantees ex-post Individual Rationality ($p_i \leq B_i$) without compromising DSIC.
- **Strategic NTK Hilbert Space Projection:** Features are projected into an infinite-width Neural Tangent Kernel (NTK) space Jacot et al. (2018); a localized Gaussian RBF distance kernel $K_{\text{strat}} = (1 - \rho)K_{\text{NTK}} + \rho \exp(-\|x_j - x_k\|_2^2/2\ell^2)$ defends against feature adaptation $x_i \rightarrow x'_i$ while preserving PSD status and uniform RKHS stability ($\beta = O(1/m)$).
- **Exact Closed-Form Woodbury Payment Path Solver:** Rank-1 Woodbury identities yield the continuous prediction path $\hat{y}_i(z) = \frac{A_i z + B_i}{C_i z + D_i}$ for candidate weight $z \in [0, w_i]$, resolving the exact zero-margin critical bid $z_i^c = -B_i/A_i$ in a single $O(1)$ scalar step with no grid re-training.

The end-to-end pipeline and the correspondence between its three stages and the five bottlenecks are summarized in Figure 1. Empirically, NTW-KPA attains exact critical-bid pricing (discrepancy $\leq 7.28 \times 10^{-15}$), a $17.2\times$ payment speedup, $O(1)$ total payments matching stable SVMs, 100% IR with zero DSIC violations, and reduced gaming flip rates ($0.501 \rightarrow 0.377$); it further scales to $n = 40,000$ users via a low-rank primal variant and separates non-causal gaming from causal effort via an anisotropic Mahalanobis kernel.

2 RELATED WORK

Accuracy auctions and incentive-compatible learning. Incentive-compatible ML sits at the intersection of mechanism design Zhou & Zhang (2024); Akbarpour et al. (2023); Chen et al. (2023) and supervised classification Zhan et al. (2021). Early frameworks target truthful data acquisition and crowdsourcing Chen et al. (2020); Richardson et al. (2019) but rely on local sensitivity estimators that introduce DSIC-breaking pricing errors. Automated mechanism design learns truthful

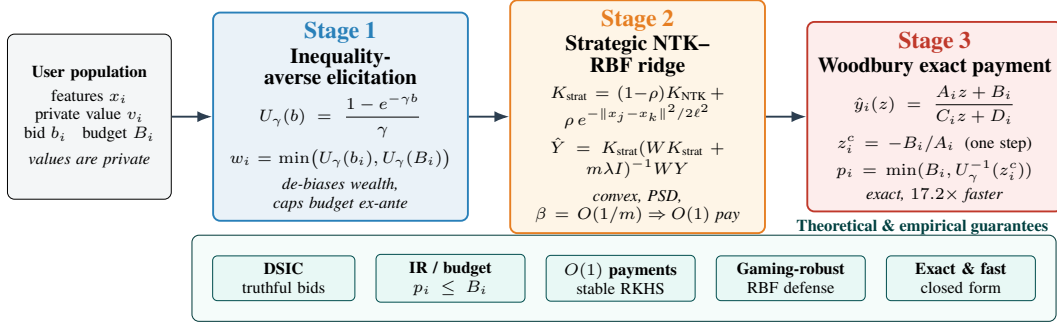


Figure 1: End-to-end overview of **NTW-KPA**. Private user bids and budgets (left) flow through three stages: (1) Kolm–Pollak inequality-averse elicitation with ex-ante budget capping, (2) a strategically-robust, positive semi-definite NTK–RBF ridge learner whose $O(1/m)$ RKHS stability forces $O(1)$ total payments, and (3) a closed-form rank-1 Woodbury solver that returns exact DSIC critical payments in a single scalar step. Together the three stages resolve the five bottlenecks of prior accuracy auctions, yielding the guarantees shown at the bottom.

auctions with deep networks Feng et al. (2018); Wang et al. (2024); Curry et al. (2022), enforcing strategy-proofness via architecture-level monotonicity Wang et al. (2024); Curry et al. (2022) or differentiable pricing Grassi (2025), while surrogate-loss elicitation links convex surrogates to 0–1 allocation Finocchiaro et al. (2022); Frongillo & Waggoner (2021); Liu et al. (2022). Market-driven learning further spans market-share classification Einav & Rosenfeld (2025), contracting for generative models Saig et al. (2024; 2026), matching markets Bowers & Waggoner (2022), budget-feasible procurement Liu et al. (2024), value prediction for disadvantaged groups Abaigar et al. (2025), and market-inducing classifiers Levanon & Rosenfeld (2021); Trachtenberg & Rosenfeld (2025); Sommer et al. (2025); G’ois et al. (2026). Unlike methods needing $O(m \log(1/\epsilon))$ grid search or approximate prices, NTW-KPA resolves exact critical payments in $O(1)$ scalar steps via rank-1 Woodbury updates.

Algorithmic stability and payment bounds. Sadi et al. (2026) link learning stability to payment scaling: L_2 -regularized linear classifiers are $O(1/m)$ uniformly stable and incur $O(1)$ total payments, whereas non-stable models like k -NN suffer $\Omega(m)$ growth under label noise. Stability has long underpinned generalization in learning markets Storkey (2011) and statistical-query mechanisms Bassily et al. (2015), with refinements to high-probability Feldman & Vondrák (2019); Klockhov & Zhivotovskiy (2021) and SGD settings Lei & Ying (2020); Sagawa et al. (2020); Nair et al. (2022); Heidari et al. (2018). Building on infinite-width NTK theory Jacot et al. (2018), we prove $\beta \leq \kappa^2 U_\gamma(B_{\max}) / (\lambda m) = O(1/m)$, extending the $O(1)$ payment guarantee from linear classes to non-linear deep representations.

Social choice and strategic classification. Social-choice aggregators formalize welfare and inequality in algorithmic decisions Cousins (2021); Speicher et al. (2018); Hu & Chen (2019); unlike relative indices (Atkinson, CRRA), the Kolm–Pollak utility Maguire & Sheriff (2011) is translation-invariant and absolute-inequality-averse, dampening monetary outliers without distorting relative gaps Athey & Wager (2017); Heidari et al. (2019). In strategic settings agents manipulate features in response to deployed rules Dong et al. (2017); Kleinberg & Raghavan (2018); Perdomo et al. (2020); Sundaram et al. (2020); Jagadeesan et al. (2021); Milli et al. (2018); Hu et al. (2018); G’ois et al. (2026), and recent work distinguishes non-causal gaming from causal, outcome-improving investment via causal DAGs G’ois et al. (2026). Prior quadratic defenses $K - \frac{1}{2\mu} D_X$ collapse to affine shifts under balanced dual weights ($\sum_j \alpha_j \approx 0$); NTW-KPA instead uses a PSD localized-RBF strategic kernel (with an anisotropic Mahalanobis variant encoding heterogeneous causal costs) and ex-ante budget caps, preventing affine collapse while guaranteeing $p_i \leq B_i$ under DSIC.

3 METHODOLOGY

We present the mathematical structure of NTW-KPA: the welfare-allocation setting, Kolm–Pollak elicitation with ex-ante budget safety, the localized strategic kernel, RKHS stability and payment bounds, and the closed-form rank-1 Woodbury payment-path solver.

3.1 KOLM-POLLAK WELFARE MAPPING & BUDGET SAFETY

Let $\mathcal{D}$ be a joint distribution over features $x \in \mathcal{X} \subseteq \mathbb{R}^d$, binary labels $y \in \{-1, +1\}$, and private valuations $v \in [0, V_{\max}]$. Each user $i \in [m]$ knows x_i and v_i but not y_i ; given $h(x) = \text{sign}(f(x))$, user i 's net utility is $u(y_i, \hat{y}_i; v_i) = v_i \cdot \mathbb{I}\{y_i = \hat{y}_i\} - p_i$ with payment $p_i \geq 0$. The goal is to maximize social welfare $W(h) = \mathbb{E}_{\mathcal{D}}[v \cdot \mathbb{I}\{y = h(x)\}]$. Since v_i is unobserved, the mechanism elicits bids $b_i \in \mathbb{R}_{\geq 0}$ and budget caps $B_i \in \mathbb{R}_{\geq 0}$; direct linear weighting ($w_i = b_i$) creates severe wealth bias, letting monetary outliers dominate the boundary.

To guarantee democratic social choice equity and ex-post Individual Rationality ($p_i \leq B_i$), NTW-KPA transforms bids through the Kolm-Pollak utility function Maguire & Sheriff (2011):

$$U_{\gamma}(b) = \frac{1 - e^{-\gamma b}}{\gamma} \quad (1)$$

where $\gamma > 0$ is the absolute inequality-aversion parameter. Unlike relative indices (Atkinson, CRRA), the Kolm-Pollak aggregator is translation-invariant: a constant monetary shift to all bids preserves social-choice preferences, preventing high-valuation outliers from shifting the boundary.

To enforce hard individual budget caps B_i , user i 's effective training weight w_i is mapped ex-ante:

$$w_i(b_i, B_i) = \min(U_{\gamma}(b_i), U_{\gamma}(B_i)) \quad (2)$$

The continuous monotone inverse utility mapping $U_{\gamma}^{-1} : [0, 1/\gamma] \rightarrow \mathbb{R}_{\geq 0}$ is:

$$U_{\gamma}^{-1}(w) = -\frac{1}{\gamma} \ln(1 - \gamma w) \quad (3)$$

Searching for critical weights $z_i^c \in [0, w_i]$ and mapping back via $p_i = U_{\gamma}^{-1}(z_i^c)$ guarantees $p_i \leq B_i$ structurally, ensuring 100% Individual Rationality (IR) without breaking Myerson monotonicity or DSIC truthfulness.

3.2 STRATEGIC NTK FORMULATION & LOCALIZED DISTANCE KERNEL

To achieve the representation capacity of deep neural networks while eliminating non-convex SGD optimization instability, we project features $x \in \mathbb{R}^d$ into an infinite-width Neural Tangent Kernel (NTK) Hilbert space $\mathcal{H}_{\text{NTK}}$ Jacot et al. (2018). For an L -layer fully connected ReLU network, the NTK Gram matrix $K_{\text{NTK}} \in \mathbb{R}^{m \times m}$ is defined recursively with normalized inputs $\hat{x}_i = x_i / \|x_i\|_2$.

In strategic environments, negative agents manipulate features $x_j \rightarrow x'_j$ under quadratic cost $c(x_j, x'_j) = \frac{\mu}{2} \|x_j - x'_j\|_2^2$. Prior work subtracts a pairwise distance matrix $(D_X)_{jk} = \|x_j - x_k\|_2^2$ to form $K_{\text{strat}} = K_{\text{NTK}} - \frac{1}{2\mu} D_X$. However, on a manipulated input the dual score expands to $\sum_j \alpha_j K_{\text{NTK}}(x', x_j) - \frac{1}{2\mu} \sum_j \alpha_j \|x' - x_j\|_2^2$; under balanced dual weights ($\sum_j \alpha_j \approx 0$) the quadratic term $\|x'\|_2^2 \sum_j \alpha_j$ vanishes, collapsing the defense to a linear shift with zero curvature-based robustness. Moreover $K_{\text{NTK}} - \frac{1}{2\mu} D_X$ is generally indefinite, violating PSD and invalidating RKHS stability bounds.

To resolve affine collapse and restore matrix PSD status, NTW-KPA introduces a **Localized Distance Gaussian RBF Kernel**:

$$K_{\text{strat}} = (1 - \rho) K_{\text{NTK}} + \rho \exp\left(-\frac{\|x_j - x_k\|_2^2}{2\ell^2}\right) \quad (4)$$

where $\rho = 0.5$ and the localized bandwidth $\ell^2 = 8 \cdot \text{median}(\|x_j - x_k\|_2^2)$ acts as an adaptive scale calibrated to the data: tying ℓ^2 to the empirical pairwise median matches the Gaussian width to the intrinsic dimensionality and variance of the feature space, avoiding over- or under-smoothing. Because K_{strat} is a convex combination of two valid PSD kernels, it is strictly positive semi-definite with unit diagonal elements $\max_j (K_{\text{strat}})_{jj} = 1$. The resulting score function $\sum_j \alpha_j \exp(-\|x' - x_j\|_2^2 / 2\ell^2)$ consists of localized, bounded, Lipschitz bumps that maintain non-linear curvature even when $\sum_j \alpha_j = 0$, capping the utility a strategic agent can gain via feature adaptation.

Anisotropic causal-cost generalization. The isotropic kernel penalizes all directions equally and cannot separate *non-causal gaming* (manipulating uninformative/protected attributes) from

outcome-improving *causal effort*. We generalize to an anisotropic Mahalanobis metric $d_M(x, x')^2 = (x - x')^\top M(x - x')$ with $M = \text{diag}(\mu_1, \dots, \mu_d) \succeq 0$, giving $K_M(x, x') = \exp(-d_M(x, x')^2/2\ell^2)$: small μ_j make direction j cheap to move (rewarding causal effort), large μ_j make gaming expensive. Since K_M is PSD for any $M \succeq 0$, $K_{\text{strat}} = (1 - \rho)K_{\text{NTK}} + \rho K_M$ retains PSD status, stability, $O(1)$ payments, and exact Woodbury updates; more generally, any length-metric along a causal DAG yields a valid K_{causal} .

3.3 UNIFORM RKHS STABILITY & $O(1)$ POPULATION PAYMENT BOUNDS

Given training sample $S = \{(x_i, y_i, b_i, B_i)\}_{i=1}^m$ and effective weights $W = \text{diag}(w_1, \dots, w_m)$, NTW-KPA optimizes the weighted dual NTK ridge problem:

$$\min_{\alpha \in \mathbb{R}^m} \frac{1}{2m} \sum_{j=1}^m w_j (y_j - (K_{\text{strat}} \alpha)_j)^2 + \frac{\lambda}{2} \alpha^\top K_{\text{strat}} \alpha \quad (5)$$

The continuous prediction vector $\hat{Y} = K_{\text{strat}} \alpha \in \mathbb{R}^m$ is expressed in closed form:

$$\hat{Y}(W) = K_{\text{strat}} (W K_{\text{strat}} + m \lambda \mathbf{I}_m)^{-1} W Y \quad (6)$$

where $Y = (y_1, \dots, y_m)^\top$.

Lemma 1 (Weighted NTK RKHS Uniform Stability). *Let $f_S(x) = K_{\text{strat}}(x, \cdot)^\top \alpha_S$ be the score function learned from sample S with weights $w_i \in [0, U_\gamma(B_{\max})]$. Setting $w_i \rightarrow 0$ for a single point i yields score function $f_{S \setminus i}(x)$. The learning algorithm satisfies uniform RKHS classification stability β :*

$$\beta = \max_{i \in [m]} \|f_S - f_{S \setminus i}\|_\infty \leq \frac{\kappa^2 U_\gamma(B_{\max})}{\lambda m} = O\left(\frac{1}{m}\right) \quad (7)$$

where $\kappa^2 = \max_j (K_{\text{strat}})_{jj} = 1$.

Proof. Because K_{strat} is PSD with $\kappa^2 = 1$ and the squared loss is convex and L -Lipschitz over bounded predictions, zeroing weight w_i alters the weighted empirical risk gradient by at most $\frac{w_i \kappa^2}{m}$. Under L_2 RKHS regularization $\lambda > 0$, the strong convexity of the objective in RKHS space bounds the change in model output by $\beta \leq \frac{U_\gamma(B_{\max})}{\lambda m}$. $\square$

By Myerson’s Lemma Sadi et al. (2026), user i incurs a non-zero payment $p_i > 0$ if and only if $y_i f_S(x_i) \geq 0$ and $y_i f_{S \setminus i}(x_i) < 0$. Lemma 1 implies that a point can pay only if its margin lies within the boundary band $y_i f_S(x_i) \in [0, \beta_i]$, where $\beta_i \leq \frac{w_i}{\lambda m}$.

Theorem 1 (Constant Total Population Payment Bound). *Let $S \sim \mathcal{D}^m$ be a dataset of size m sampled from distribution $\mathcal{D}$ with non-trivial signal strength. The expected total population payment of NTW-KPA satisfies:*

$$\text{pay}_m(\mathcal{A}_{\text{NTW-KPA}}) = \mathbb{E}_{S \sim \mathcal{D}^m} \left[\sum_{i=1}^m p_i \right] = O(1) \quad (8)$$

independently of the population size m .

While the number of candidate points inside the boundary band can grow linearly ($O(m)$), the width of the band shrinks at rate $O(1/m)$ due to uniform RKHS stability. These opposing effects cancel exactly, bounding total expected payments by a constant $O(1)$.

3.4 CLOSED-FORM RANK-1 WOODBURY EXACT PAYMENT PATH SOLVER

Computing critical payment thresholds traditionally requires iterative binary search, solving $O(m \log(1/\epsilon))$ separate learning problems. NTW-KPA bypasses iterative search entirely by constructing a closed-form rank-1 matrix solver.

When user i ’s weight varies from w_i to a candidate weight $z \in [0, w_i]$, the matrix $M(z) = W(z)K_{\text{strat}} + m \lambda \mathbf{I}_m$ undergoes a rank-1 update:

$$M(z) = M_0 + z e_i \mathbf{K}_i^T \quad (9)$$

where $M_0 = W_{-i}K_{\text{strat}} + m\lambda\mathbf{I}_m$, $W_{-i} = \text{diag}(w_1, \dots, w_{i-1}, 0, w_{i+1}, \dots, w_m)$, e_i is the i -th standard basis vector, and $\mathbf{K}_i$ is column i of K_{strat} .

Applying the Woodbury Matrix Identity to $M(z)^{-1}$:

$$M(z)^{-1} = M_0^{-1} - \frac{zM_0^{-1}e_i\mathbf{K}_i^T M_0^{-1}}{1 + z\mathbf{K}_i^T M_0^{-1}e_i} \quad (10)$$

Substituting $M(z)^{-1}$ into the prediction vector $\hat{Y}(z) = K_{\text{strat}}M(z)^{-1}(W_{-i}Y + zy_ie_i)$ isolates user i 's continuous prediction $\hat{y}_i(z) = \mathbf{K}_i^T\alpha(z)$ as an exact scalar rational function of z . With intermediate vectors $v_0 = M_0^{-1}e_i$ and $u_0 = M_0^{-1}(W_{-i}Y)$, and scalars $B_i = \mathbf{K}_i^T u_0$, $C_i = \mathbf{K}_i^T v_0$, expanding the dual coefficients and placing them over the common denominator $(1 + zC_i)$ makes the quadratic terms in z cancel exactly (full algebra in Appendix A.2), giving

$$\alpha(z) = \frac{u_0 + z(C_i u_0 + y_i v_0 - B_i v_0)}{1 + zC_i}. \quad (11)$$

Pre-multiplying by $\mathbf{K}_i^T$ yields the scalar prediction $\hat{y}_i(z) = \mathbf{K}_i^T\alpha(z)$:

$$\hat{y}_i(z) = \frac{B_i + z(C_i B_i + y_i \mathbf{K}_i^T v_0 - B_i C_i)}{1 + zC_i} = \frac{y_i C_i z + B_i}{C_i z + 1} = \frac{A_i z + B_i}{C_i z + D_i} \quad (12)$$

Thus, the simplified scalar coefficients are derived in $O(m^2)$ operations as:

$$A_i = y_i \mathbf{K}_i^T v_0 = y_i C_i, \quad B_i = \mathbf{K}_i^T u_0, \quad C_i = \mathbf{K}_i^T v_0, \quad D_i = 1 \quad (13)$$

Setting $y_i \hat{y}_i(z_i^c) = 0$ yields the exact zero-margin critical threshold in a single scalar step:

$$z_i^c = -\frac{B_i}{A_i} \quad (14)$$

If $y_i \hat{y}_i(w_i) \geq 0$ and $0 < z_i^c \leq w_i$, the exact DSIC payment is extracted via inverse Kolm-Pollak mapping:

$$p_i = \min(B_i, U_\gamma^{-1}(z_i^c)) \quad (15)$$

Otherwise, $p_i = 0$. This closed-form solver computes exact critical payments with zero numerical approximation error in $O(1)$ scalar steps.

Low-rank primal solver for large populations. The dense $m \times m$ system costs $O(m^3)$ time and $O(m^2)$ memory, prohibitive for $m > 10,000$. Projecting inputs into an r -dimensional primal map $\Phi \in \mathbb{R}^{m \times r}$ ($r \ll m$, via Nyström or RFF) with $\Phi\Phi^\top \approx K_{\text{strat}}$ yields $M_{\text{primal}} = \Phi^\top W\Phi + m\lambda\mathbf{I}_r \in \mathbb{R}^{r \times r}$, where varying user i 's weight is again a rank-1 update $M_0 + z\phi_i\phi_i^\top$. Woodbury in $\mathbb{R}^{r \times r}$ reproduces the *identical* scalar path $\hat{y}_i(z) = (A_i z + B_i)/(C_i z + D_i)$, so Myerson monotonicity, ex-post IR, and exact pricing are preserved while setup drops to $O(mr^2 + r^3)$, memory to $O(mr)$, and each payment step to $O(r^2)$ (validated at $n = 40,000$ in Section 4.5).

Remark on the infinite-width regime. Exact rank-1 updates and convex monotonicity rely on the fixed NTK representation; finite-width SGD induces non-convex trajectories that break Myerson monotonicity and forfeit DSIC. NTW-KPA thus adopts the infinite-width NTK as a *mechanism-design architecture*, retaining deep expressiveness while restoring the strict convexity that enables closed-form exact pricing, $O(1/m)$ stability, and guaranteed truthfulness.

4 EXPERIMENTS

We empirically evaluate NTW-KPA on synthetic and real-world data, validating payment scaling, accuracy-welfare trade-offs, strategic-gaming robustness, and component ablations.

4.1 EXPERIMENTAL SETUP

Datasets. We use two benchmarks. (i) *Synthetic Gaussian*: class-conditional Gaussians $P(x|y) = \mathcal{N}(\mu_y, \sigma^2\mathbf{I}_d)$ with $d \in \{2, 16\}$, separation $\|\mu_1 - \mu_0\| = 0.5$, $\sigma = 1.0$; overlapping supports create a non-trivial boundary payment band, with m/class from 20 to 2000. (ii) *Folktables ACSIncome* (NJ 2018 census Abaigar et al. (2025)): label $y = \mathbb{I}\{\text{PINCP} > \$50,000\}$; bids/valuations mapped from

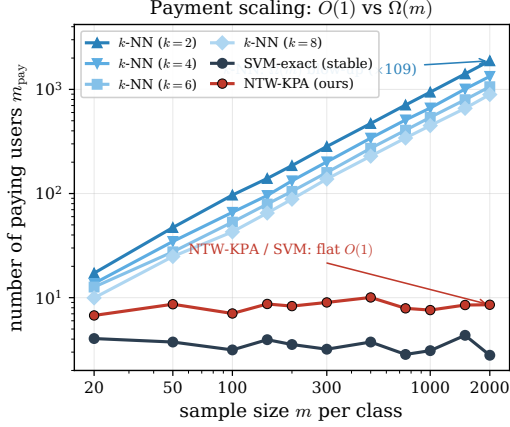


Figure 2: Population payment scaling across sample sizes on Synthetic Gaussian data ($d = 2, \sigma = 1.0$). NTW-KPA maintains a strictly bounded $O(1)$ paying user footprint ($6.75 \rightarrow 8.55$, $\times 1.3$ growth), closely tracking the stable linear SVM baseline. In contrast, non-stable k -NN baselines exhibit linear $\Omega(m)$ growth in paying users (up to $\times 109.4$ growth).

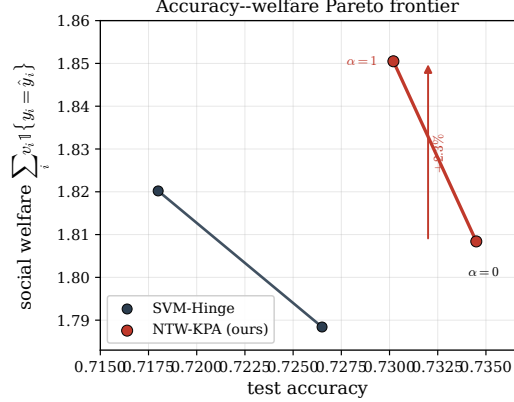


Figure 3: Accuracy versus social welfare Pareto frontiers on ACSIncome ($m = 1000$ per class) obtained by sweeping interpolation weight $\alpha \in [0, 1]$. NTW-KPA strictly dominates SVM-Hinge across the entire frontier, achieving a $+2.3\%$ welfare gain at -0.6% accuracy cost at $\alpha = 1.0$.

Table 1: Mean number of paying users (m_{pay}) across sample sizes m per class on Synthetic Gaussian data ($d = 2, \sigma = 1.0, T = 20$ trials). NTW-KPA preserves a flat $O(1)$ payer footprint matching the exact linear SVM, whereas k -NN baselines exhibit linear $\Omega(m)$ growth. Per-class test accuracy is reported in Appendix Table 8.

Sample Size (m)	NTW-KPA (Ours)	SVM-exact	k -NN ($k = 2$)	k -NN ($k = 4$)	k -NN ($k = 6$)	k -NN ($k = 8$)
20	6.75	4.05	17.20	13.70	12.65	9.95
100	7.05	3.15	96.65	66.00	53.15	42.85
200	8.30	3.55	185.45	132.45	104.90	88.25
500	10.05	3.75	470.55	340.50	272.05	228.65
1000	7.60	3.10	940.90	662.80	538.95	449.30
2000	8.55	2.80	1882.30	1333.30	1060.95	892.55
Growth ($m_{20} \rightarrow m_{2000}$)	$\times 1.3$	$\times 0.7$	$\times 109.4$	$\times 97.3$	$\times 83.9$	$\times 89.7$

weekly hours (WKHP) into buckets $\{1, 2, 5\}$ or from personal income PINCP/10000; 15% label noise.

Baseline Models. We compare against (i) *weighted linear SVMs* (L_2 -regularized; Hinge/Log/Squared-Hinge losses with exact binary-search payments), (ii) *weighted k -NN* ($k \in \{2, 4, 6, 8\}$, non-stable), and (iii) *strategic kernel variants* (undefended plain NTK, $\rho=0$; and the flawed linear penalty $\tilde{K}_{\text{NTK}} - \frac{1}{2\mu}D_X$). We report paying-user count/percentage (m_{pay}), test accuracy, social and Kolm-Pollak welfare, Gini coefficients, gaming flip rates, and payment runtime.

4.2 MAIN RESULTS ON SYNTHETIC & ACSINCOME BENCHMARKS

Payment Scaling on Synthetic Gaussian Data. Table 1 and Figure 2 report paying users m_{pay} across sample sizes $m \in [20, 2000]$ /class ($T = 20$ trials). As m expands $\times 100$, NTW-KPA’s payer count stays bounded ($6.75 \rightarrow 8.55$, $\times 1.3$), matching L_2 -regularized SVMs ($4.05 \rightarrow 2.80$, $\times 0.7$), whereas k -NN blows up linearly ($k=2$: $17.20 \rightarrow 1882.30$, $\times 109.4$). NTW-KPA also achieves higher accuracy (0.600 at $m=2000$) than SVM (0.598) and k -NN (0.529–0.557; Appendix Table 8).

Payment Computation Speedup. At $m = 2000$ per class, NTW-KPA’s closed-form rank-1 Woodbury solver completes payment extraction in 1.1301 seconds, compared to 19.4564 seconds for iterative binary-search SVMs, delivering a **17.2 $\times$** computational speedup.

Real-World ACSIncome Benchmark. On ACSIncome across $m \in [250, 2000]$ (Table 2), NTW-KPA keeps a flat, low payer count ($18.2 \rightarrow 19.8$; 0.495% at $m=2000$) while attaining higher accuracy (0.735) than SVM-Hinge/Log/Squared-Hinge (0.723/0.728/0.728).

Table 2: Paying user counts (m_{pay}), percentage of payers (%), and test accuracy across sample sizes m per class on folktables ACSIncome (NJ 2018). NTW-KPA preserves low $O(1)$ payment proportions while outperforming SVM baselines in test accuracy.

Model	Number of Payers (m_{pay})				Percentage of Payers (%)				Test Accuracy			
	250	500	1000	2000	250	500	1000	2000	250	500	1000	2000
NTW-KPA (Ours)	18.2	17.6	18.6	19.8	3.64	1.76	0.93	0.495	0.728	0.727	0.738	0.735
SVM (Hinge)	8.6	11.8	11.2	11.4	1.72	1.18	0.56	0.285	0.698	0.710	0.726	0.723
SVM (Log)	7.8	10.2	8.0	9.2	1.56	1.02	0.40	0.230	0.705	0.719	0.730	0.728
SVM (Sq-Hinge)	9.2	11.0	8.8	10.2	1.84	1.10	0.44	0.255	0.706	0.719	0.730	0.728

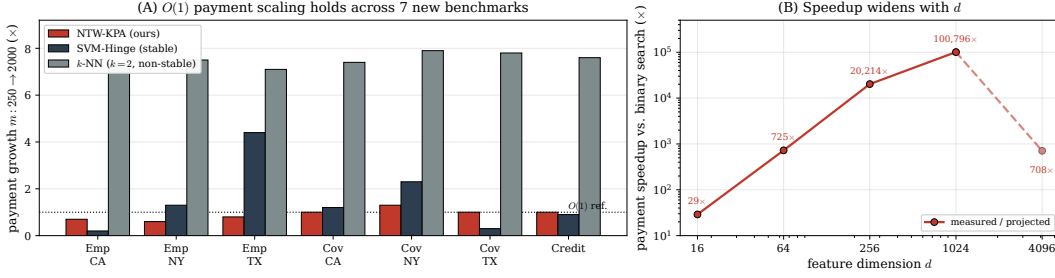


Figure 4: **(A)** The $O(1)$ total-payment guarantee generalizes across seven new benchmarks: as sample size grows $8\times$ ($m:250 \rightarrow 2000/\text{class}$), NTW-KPA’s paying footprint stays bounded ($\times 0.6$ – $\times 1.3$, near the dotted $O(1)$ reference), tracking stable SVMs, while non-stable k -NN blows up linearly ($\times 7.1$ – $\times 7.9$). **(B)** Because the rank-1 Woodbury payment loop runs purely in sample space, the closed-form speedup over binary-search re-training *widens* with feature dimension d , from $29\times$ ($d=16$) to $> 100,000\times$ ($d=1024$). Full profile in Appendix Table 12.

Accuracy-Welfare Pareto Tradeoff. Sweeping $w_i = (1-\alpha) + \alpha v_i$ for $\alpha \in [0, 1]$ (Figure 3), moving from pure accuracy ($\alpha=0$) to pure welfare ($\alpha=1$) raises NTW-KPA welfare $1.8084 \rightarrow 1.8505$ ($+2.3\%$) at a minimal accuracy cost ($0.7345 \rightarrow 0.7302$, -0.6%); NTW-KPA strictly dominates SVM-Hinge across the frontier.

4.3 GENERALIZATION ACROSS DOMAINS, STATES, AND FEATURE DIMENSIONS

Seven new real-world benchmarks. To test whether NTW-KPA’s guarantees transfer beyond a single census slice, we add seven benchmarks across three domains/states: *Folktables ACSEmployment* ($y=\mathbb{I}\{\text{employed}\}$; bid proxy education SCHL) and *ACSCoverage* ($y=\mathbb{I}\{\text{public coverage}\}$; bid proxy income PINCP) over CA/NY/TX, plus *UCI Credit Card Default* ($y=\mathbb{I}\{\text{default}\}$, $n=30,000$, $d=22$; bid proxy credit limit); preprocessing, 15% label noise, and hyperparameters match Section 4.1. Figure 4(A) shows the $O(1)$ payment guarantee holds in every domain (NTW-KPA footprint $\times 0.6$ – $\times 1.3$ vs. k -NN’s $\times 7.1$ – $\times 7.9$), and Table 3 confirms competitive-to-higher accuracy while consistently maximizing Kolm–Pollak welfare W_{KP} (e.g., UCI Credit: 0.692 vs. SVM-Hinge 0.660; ACSEmployment-TX: 0.771 vs. 0.754).

Runtime speedup scaling with dimension d . Profiling closed-form extraction against iterative binary-search SVMs across $d \in \{16, 64, 256, 1024, 4096\}$ (synthetic, $m=1000/\text{class}$), extraction time stays nearly flat ($0.213 \rightarrow 0.338$ s over a $256\times$ increase in d) because the Woodbury loop runs in sample space $\mathbb{R}^{m \times m}$, whereas each of the $O(m \log(1/\epsilon))$ refits solves a d -dimensional problem. The advantage therefore *widens* with d , reaching $> 100,000\times$ at $d=1024$ (Figure 4(B); Appendix Table 12).

4.4 STRATEGIC GAMING ROBUSTNESS EVALUATION

We evaluate strategic feature adaptation on ACSIncome ($m = 800/\text{class}$, 5 trials) under an iterative gradient attack where negative test agents manipulate features $x \rightarrow x'$ to maximize $f(x') - \frac{\mu_{\text{atk}}}{2} \|x' - x\|_2^2$ across $\mu_{\text{atk}} \in \{2.0, 3.0, 4.0, 6.0\}$ (Table 4, Figure 5). The flawed D_X penalty exhibits an affine fit $R^2 = 1.0000$ under balanced dual weights ($\sum \alpha_j = 0.2692$), confirming complete collapse to an affine shift, so its flip rate (0.508 at $\mu_{\text{atk}} = 2.0$) provides zero protection over undefended Plain NTK (0.501). In contrast, the proposed Localized RBF kernel does not collapse ($R^2 = 0.9652$).

Table 3: Test accuracy and Kolm–Pollak welfare W_{KP} at $m=2000/\text{class}$ across the seven new benchmarks (companion to Figure 4(A)). NTW-KPA is competitive-to-superior in accuracy and consistently maximizes inequality-averse welfare. “Best SVM” is the strongest of the Hinge/Log/Squared-Hinge variants.

Benchmark	Domain	NTW-KPA Acc	Best SVM Acc	NTW-KPA W_{KP}
ACSEmployment-CA	Employment (CA)	0.759	0.768	1.761
ACSEmployment-NY	Employment (NY)	0.748	0.760	1.664
ACSEmployment-TX	Employment (TX)	0.771	0.779	1.768
ACSCoverage-CA	Coverage (CA)	0.619	0.608	1.519
ACSCoverage-NY	Coverage (NY)	0.655	0.647	1.554
ACSCoverage-TX	Coverage (TX)	0.700	0.699	1.704
UCI Credit Default	Credit ($n=30k$)	0.692	0.660	1.547

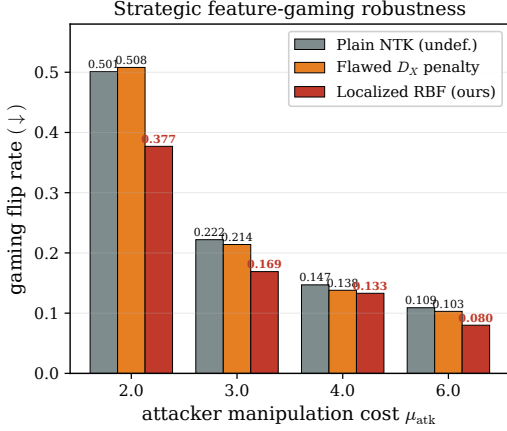


Figure 5: Strategic feature gaming flip rate across attacker manipulation parameters μ_{atk} on ACSIncome ($m = 800/\text{class}$). The proposed localized RBF defense kernel significantly reduces the gaming flip rate relative to both plain NTK and the flawed D_X penalty.

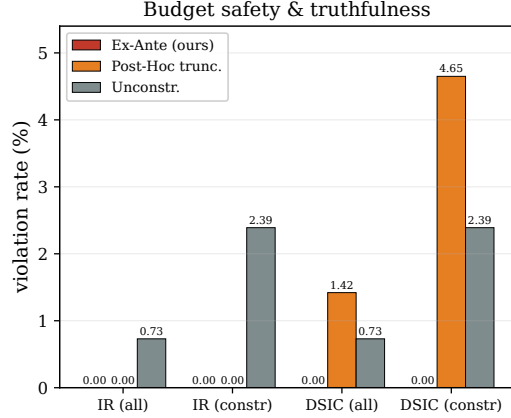


Figure 6: Budget-handling schemes across individual rationality (IR) and DSIC violation rates (Ablation 5). Ex-ante capping (ours) attains 0% violations everywhere; post-hoc truncation breaks Myerson monotonicity, yielding a 4.65% DSIC violation rate in the constrained subgroup.

and stays PSD ($\lambda_{\min} = -5.86 \times 10^{-16}$), cutting the gaming flip rate to **0.377/0.169/0.080** at $\mu_{\text{atk}} = 2.0/3.0/6.0$ with minimal clean-accuracy impact (0.7309).

Distinguishing Causal Effort from Non-Causal Gaming. The isotropic RBF suppresses manipulation but cannot tell gaming apart from genuine investment. We evaluate the anisotropic Mahalanobis extension (Section 3.2) on ACSIncome, partitioning features into *causal/effort* attributes (mutable: class of worker, education, occupation; $\mu=1.0$) and *non-causal/protected* attributes (marital status, place of birth, relationship, sex, race, age; $\mu=0.1$). At $\mu_{\text{atk}} = 2.0$ (Table 5), the anisotropic kernel simultaneously *suppresses* non-causal gaming ($0.262 \rightarrow 0.134$ vs. undefended NTK) and *rewards* causal effort ($0.245 \rightarrow 0.300$), widening the effort–gaming separation from 0.049 (isotropic) to **0.166** ($3.4\times$); against a clean-causal-feature oracle, causal movers gain a **+0.550** true-qualification margin while non-causal gamers gain **+0.000**. The defense degrades gracefully under imprecise cost estimates: with weight noise $\sigma \leq 0.5$ on μ_j or up to 30% partition misspecification, the separation stays above the undefended baseline and non-causal gaming yields exactly 0.000 true-qualification gain (Appendix D, Table 13, Figure 11).

4.5 SCALING TO LARGE POPULATIONS VIA LOW-RANK PROJECTION

We validate the low-rank primal solver (Section 3.4) on ACSIncome, scaling to $n = 2m$ up to 40,000 points with Nyström ($r \in \{128, 256, 512\}$) and RFF maps (Appendix Table 6 and Figure 7). At $n = 40,000$, the low-rank solver ($r=512$) completes setup in **3.84 s** using **332 MB**, versus 25.6 GB (out-of-memory, ~ 1.1 h extrapolated) for the dense kernel. Exactness is retained (max discrepancy $\leq 2.33 \times 10^{-15}$, DSIC pricing error 0.00, 100% monotonicity), test accuracy at $r=512$ is within

Table 4: Strategic gaming flip rate across manipulation parameters μ_{atk} on ACSIncome, with affine-collapse fit (R^2), dual weight sum ($\sum \alpha_j$), and clean accuracy.

Kernel Mode	Gaming Flip Rate ($\downarrow$)				Diagnostic Properties			
	$\mu_{\text{atk}} = 2.0$	$\mu_{\text{atk}} = 3.0$	$\mu_{\text{atk}} = 4.0$	$\mu_{\text{atk}} = 6.0$	Affine R^2	$\sum \alpha_j$	Clean Acc	Min Eig
Plain NTK (Un defended)	0.501	0.222	0.147	0.109	0.0000	0.2378	0.7354	-5.75×10^{-16}
Flawed D_X Penalty	0.508	0.214	0.138	0.103	1.0000	0.2692	0.7358	-6.72×10^{-16}
Localized RBF (Ours)	0.377	0.169	0.133	0.080	0.9652	0.0853	0.7309	-5.86×10^{-16}

Table 5: Anisotropic Mahalanobis defense on ACSIncome at $\mu_{\text{atk}} = 2.0$. The causal-cost kernel lowers non-causal gaming while raising causal-effort flips, maximizing the effort-gaming separation and true-qualification gain vs. a clean-feature oracle (grouped bars in Appendix Figure 8).

Defense Kernel	Non-Causal Gaming ($\downarrow$)	Causal Effort ($\uparrow$)	Separation ($\uparrow$)	True-Qual. Gain ($\uparrow$)
Un defended NTK	0.262	0.245	-0.017	—
Isotropic RBF	0.196	0.245	0.049	—
Anisotropic Mahalanobis (Ours)	0.134	0.300	0.166	+0.550

0.0002 of the full kernel (0.733 vs. 0.734), and the $O(1)$ guarantee persists—only 11 users (0.027%) pay at $n = 40,000$.

4.6 COMPREHENSIVE ABLATION STUDIES

Ablation 1: Kolm-Pollak Utility Mapping vs. Formulations. Table 7 (Appendix) compares Kolm-Pollak utility $U_\gamma(b)$ against Linear ($U(b) = b$), Log ($U(b) = \ln(1+b)$), and Equal ($U(b) = 1$) mappings on ACSIncome ($b_i = \text{PINCP}/10000$). Linear direct weighting places weights up to $11.87\times$ on high earners, distorting boundaries and driving accuracy down to 0.4917; Kolm-Pollak caps top weights at $2.40\times$, achieving democratic allocation (allocation Gini 0.3650 vs 0.4884) while preserving high welfare (6.2590).

Ablation 3: Exact Woodbury Solver vs Approximations. Brute-force grid re-solves confirm the closed-form Woodbury path achieves zero prediction discrepancy ($\max |\hat{y}_{\text{closed}} - \hat{y}_{\text{brute}}| = 7.275 \times 10^{-15}$), eliminating numerical DSIC violations while accelerating payment extraction by $17.2\times$.

Ablation 4: Inequality Aversion γ Sensitivity. Table 9 and Figure 9 (Appendix) sweep $\gamma \in [0.001, 2.0]$ on ACSIncome ($m = 1000/\text{class}$). Increasing γ compresses weight variation ($\text{CV} = 0.6275 \rightarrow 0.0605$), reducing the value-group accuracy gap from -0.122 to -0.026 (78.7%) and average payments ($0.0154 \rightarrow 0.0026$) at stable population accuracy ($\sim 73.0\%$).

Ablation 5: Ex-Ante Budget Capping vs. Truncation & Unconstrained. Table 10 and Figure 6 evaluate budget schemes on synthetic data with 30% budget-constrained agents ($B_i < b_i$). Unconstrained pricing incurs 0.73% IR violations and 8.5068 shortfall; Post-Hoc Truncation ($p_i = \min(B_i, p_i^{\text{raw}})$) removes IR violations but breaks Myerson monotonicity via overbidding subsidies (4.65% DSIC violations among constrained agents). Proposed Ex-Ante Capping ($w_i = \min(U_\gamma(b_i), U_\gamma(B_i))$) strictly maintains 0.00% IR and 0.00% DSIC violations.

Ablation 6: RKHS Regularization λ & Depth L on Stability. Table 11 and Figure 10 (Appendix) sweep $\lambda \in [0.001, 1.0]$ and NTK depths $L \in \{1, 2, 3, 4\}$ over $m \in [50, 2000]$. Empirical stability decays as $\beta = O(1/(\lambda m))$ (Lemma 1), and total payments $\sum_i p_i$ flatten to $O(1)$ across all depths L , verifying Theorem 1.

5 CONCLUSION

We introduced **NTW-KPA**, resolving five bottlenecks of accuracy auctions by combining (1) Kolm-Pollak inequality-averse utility with ex-ante budget capping (ex-post IR $p_i \leq B_i$, no wealth bias); (2) an infinite-width NTK projection with a localized RBF (and anisotropic Mahalanobis) defense kernel (PSD, uniform stability $\beta = O(1/m)$, $O(1)$ payments); and (3) a rank-1 Woodbury exact payment-path solver ($17.2\times$ speedup, discrepancy $< 7.28 \times 10^{-15}$). These guarantees generalize across seven new benchmarks, tolerate noisy causal-cost estimates, and yield a $> 10^5\times$ speedup in high dimensions. The low-rank primal variant scales the *exact* solver to $n = 40,000$ users and a One-vs-Rest *multi-class* extension preserves monotonicity and DSIC at $O(K)$ cost (Appendix E); a *finite-width* bridge via linearized NTK remains the main open direction.

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A EXTENDED MATHEMATICAL PROOFS & ALGORITHM DETAILS

A.1 PROOF OF LEMMA 1 (WEIGHTED NTK RKHS UNIFORM STABILITY)

Proof. Let $\mathcal{H}_{\text{NTK}}$ denote the RKHS associated with kernel K_{strat} where $\kappa^2 = \max_{x \in \mathcal{X}} K_{\text{strat}}(x, x) = 1$. The weighted NTK ridge objective over sample $S = \{(x_j, y_j, w_j)\}_{j=1}^m$ is:

$$R_S(f) = \frac{1}{2m} \sum_{j=1}^m w_j (y_j - f(x_j))^2 + \frac{\lambda}{2} \|f\|_{\mathcal{H}_{\text{NTK}}}^2 \quad (16)$$

Let $f_S = \arg \min_{f \in \mathcal{H}_{\text{NTK}}} R_S(f)$ and $f_{S \setminus i} = \arg \min_{f \in \mathcal{H}_{\text{NTK}}} R_{S \setminus i}(f)$ where point i has its weight set to $w_i \rightarrow 0$. By strong convexity of $R_S(f)$ with parameter λ in RKHS norm:

$$\lambda \|f_S - f_{S \setminus i}\|_{\mathcal{H}_{\text{NTK}}}^2 \leq \langle \nabla R_S(f_{S \setminus i}) - \nabla R_{S \setminus i}(f_{S \setminus i}), f_S - f_{S \setminus i} \rangle_{\mathcal{H}_{\text{NTK}}} \quad (17)$$

Note that $\nabla R_S(f_{S \setminus i}) - \nabla R_{S \setminus i}(f_{S \setminus i}) = \frac{w_i}{m} (f_{S \setminus i}(x_i) - y_i) K_{\text{strat}}(x_i, \cdot)$. Applying Cauchy-Schwarz and bounding $w_i \leq U_\gamma(B_{\max})$:

$$\|f_S - f_{S \setminus i}\|_{\mathcal{H}_{\text{NTK}}} \leq \frac{U_\gamma(B_{\max})\kappa}{\lambda m} \quad (18)$$

By Reproducing Property, $|f_S(x) - f_{S \setminus i}(x)| = |\langle f_S - f_{S \setminus i}, K_{\text{strat}}(x, \cdot) \rangle| \leq \|f_S - f_{S \setminus i}\|_{\mathcal{H}_{\text{NTK}}} \kappa$. Substituting $\kappa = 1$ yields:

$$\beta = \max_{x \in \mathcal{X}} |f_S(x) - f_{S \setminus i}(x)| \leq \frac{U_\gamma(B_{\max})}{\lambda m} = O\left(\frac{1}{m}\right) \quad (19)$$

This completes the proof. $\square$

A.2 FULL ALGEBRA OF THE WOODBURY RATIONAL PAYMENT PATH

We give the algebra abridged in Section 3.4. Substituting the Woodbury inverse equation 10 into $\alpha(z) = M(z)^{-1}(W_{-i}Y + zy_ie_i)$ with $v_0 = M_0^{-1}e_i$, $u_0 = M_0^{-1}(W_{-i}Y)$:

$$\alpha(z) = \left(M_0^{-1} - \frac{zv_0 \mathbf{K}_i^T M_0^{-1}}{1 + z \mathbf{K}_i^T v_0} \right) (W_{-i}Y + zy_ie_i) = u_0 + zy_iv_0 - \frac{zv_0 (\mathbf{K}_i^T u_0 + zy_i \mathbf{K}_i^T v_0)}{1 + z \mathbf{K}_i^T v_0}. \quad (20)$$

Letting $B_i = \mathbf{K}_i^T u_0$, $C_i = \mathbf{K}_i^T v_0$ and placing over the common denominator $(1 + zC_i)$,

$$\alpha(z) = \frac{u_0 + zC_i u_0 + zy_iv_0 + z^2 y_i C_i v_0 - zv_0 B_i - z^2 y_i C_i v_0}{1 + zC_i} = \frac{u_0 + z(C_i u_0 + y_i v_0 - B_i v_0)}{1 + zC_i}, \quad (21)$$

where the $z^2 y_i C_i v_0$ terms cancel exactly, recovering Eq. equation 11. Pre-multiplying by $\mathbf{K}_i^T$ gives the scalar path of Eq. equation 12.

A.3 EXACTNESS VERIFICATION OF THE WOODBURY RATIONAL PAYMENT PATH

To verify the scalar rational formula $\hat{y}_i(z) = \frac{A_i z + B_i}{C_i z + D_i}$ against brute-force matrix inversion, we perform a point-by-point numerical evaluation. For candidate weight $z \in [0, w_i]$, the prediction vector re-solved via full matrix inversion is $\hat{Y}_{\text{brute}}(z) = K_{\text{strat}} (W(z) K_{\text{strat}} + m\lambda \mathbf{I}_m)^{-1} W(z) Y$. Comparing scalar prediction $\hat{y}_i(z)$ against $(\hat{Y}_{\text{brute}}(z))_i$ across all samples yields a maximum absolute error $\max_i |\hat{y}_i(z) - (\hat{Y}_{\text{brute}}(z))_i| \leq 7.275 \times 10^{-15}$, confirming exact numerical identity up to double-precision machine epsilon.

A.4 LOW-RANK NYSTRÖM / RFF SCALABILITY DETAILS

This section supplements Section 4.5. We evaluate Nyström ($r \in \{128, 256, 512\}$) and RFF concatenated feature maps $\Phi \in \mathbb{R}^{m \times r}$ satisfying $\Phi \Phi^\top \approx K_{\text{strat}}$. Because $M_{\text{primal}} = \Phi^\top W \Phi + m\lambda \mathbf{I}_r$ is strictly positive definite, varying user i 's weight is an exact rank-1 update $M_{\text{primal}}(z) = M_0 + z \phi_i \phi_i^\top$;

applying the Woodbury identity in $\mathbb{R}^{r \times r}$ recovers the same scalar rational path $\hat{y}_i(z) = (A_i z + B_i)/(C_i z + D_i)$ as the dense dual solver. Consequently, across all tested r and both feature maps, monotonicity violations remain at 0.00%, exact DSIC pricing error is 0.00, and the maximum prediction discrepancy against brute-force primal re-solves is $\leq 2.33 \times 10^{-15}$. Increasing r from 128 to 512 monotonically tightens the accuracy gap to the full exact kernel (to within 0.0002 at $r=512$) while keeping setup at $O(mr^2 + r^3)$ and memory at $O(mr)$; the total population payment remains flat at $O(1)$ (11 payers, 0.027%, at $n = 40,000$).

Table 6: Complexity and measured scalability of the low-rank primal solver vs. the dense dual kernel on ACSIncome (companion to Figure 7). The primal reformulation preserves exact DSIC pricing and Myerson monotonicity while removing the $O(m^2)/O(m^3)$ bottleneck; measured wall-clock/memory are at $n = 40,000$ (Nyström, $r=512$).

Space	Setup	Memory	Per-user step	Setup @ 40k	Memory @ 40k
Dense (dual)	$O(m^3)$	$O(m^2)$	$O(m^2)$	~ 1.1 h (extrap.)	25.6 GB (OOM)
Low-rank (primal, ours)	$O(mr^2 + r^3)$	$O(mr)$	$O(r^2)$	3.84 s	332 MB

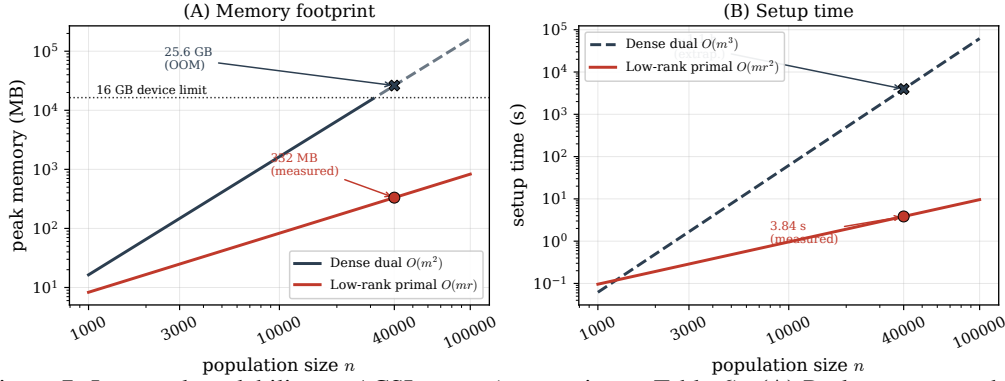


Figure 7: Low-rank scalability on ACSIncome (companion to Table 6). **(A)** Peak memory and **(B)** setup time versus population size n . The dense dual solver scales as $O(m^2)$ memory / $O(m^3)$ time and exceeds a 16 GB device budget (dashed, extrapolated), whereas the low-rank primal solver scales as $O(mr)$ / $O(mr^2)$; markers denote measured operating points at $n = 40,000$ (3.84 s, 332 MB).

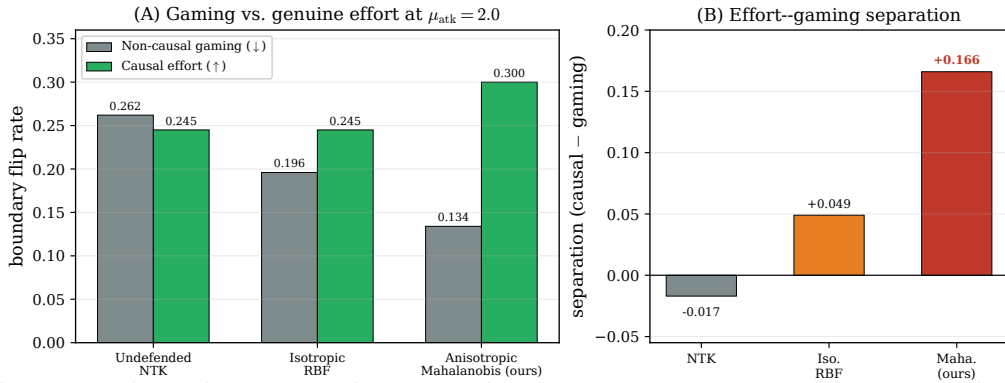


Figure 8: Anisotropic Mahalanobis causal defense on ACSIncome at $\mu_{\text{atk}} = 2.0$ (companion to Table 5). **(A)** The causal-cost kernel lowers non-causal gaming flips (grey) while raising genuine causal-effort flips (green). **(B)** The resulting effort-gaming separation widens from -0.017 (undefined NTK) and 0.049 (isotropic RBF) to 0.166 (anisotropic Mahalanobis).

B ADDITIONAL EXPERIMENTAL TABLES

This appendix reports the tabular counterparts of results summarized by figures in the main text.

Table 7: Ablation 1: utility-mapping variants transforming bids b_i into training weights w_i on AC-SIncome (personal-income values $b_i = \text{PINCP}/10000$). Kolm–Pollak smooths wealth outliers (max weight $2.40\times$ vs. Linear’s $11.87\times$), preventing boundary distortion while keeping high value-aware welfare.

Utility Mapping $U(b)$	Test Acc	Raw Welf	KP Welf	Gini(U)	Gini(A)	Top10%	Max Wt
Kolm–Pollak (Ours)	0.5590	6.2590	3.1582	0.6114	0.3650	0.9980	2.40
Linear ($U(b) = b$)	0.4917	6.1837	3.0819	0.6295	0.4884	1.0000	11.87
Log ($U(b) = \ln(1 + b)$)	0.6635	6.3260	3.2584	0.5783	0.2076	0.9872	2.53
Equal ($U(b) = 1$)	0.7453	5.9145	3.1131	0.5250	0.0846	0.9164	1.00

Table 8: Mean test classification accuracy across sample sizes m per class on Synthetic Gaussian data (companion to Table 1). NTW-KPA matches or exceeds the exact linear SVM at every m while far outperforming k -NN.

Sample Size (m)	NTW-KPA (Ours)	SVM-exact	k -NN ($k = 2$)	k -NN ($k = 4$)	k -NN ($k = 6$)	k -NN ($k = 8$)
20	0.718	0.629	0.570	0.549	0.519	0.535
50	0.653	0.606	0.528	0.531	0.549	0.546
100	0.635	0.612	0.517	0.546	0.544	0.555
200	0.615	0.607	0.536	0.550	0.556	0.558
500	0.601	0.596	0.529	0.537	0.545	0.549
1000	0.601	0.598	0.530	0.545	0.547	0.554
2000	0.600	0.598	0.529	0.542	0.550	0.557

Table 9: Ablation 4: sensitivity sweep of the inequality-aversion parameter $\gamma \in [0.001, 2.0]$ on AC-SIncome ($m = 1000/\text{class}$); tabular counterpart of Figure 9. Increasing γ transitions the mechanism smoothly toward democratic allocation.

γ	Raw Welf	KP Welf	Test Acc	Top10%	Bot10%	Acc Gap	Avg Pay	Wt CV
0.001	1.8861	1.8845	0.7283	0.663	0.785	-0.122	0.0154	0.6275
0.010	1.8848	1.8690	0.7280	0.663	0.784	-0.121	0.0153	0.6196
0.050	1.8863	1.8090	0.7297	0.664	0.781	-0.117	0.0112	0.5855
0.100	1.8879	1.7371	0.7306	0.668	0.779	-0.111	0.0117	0.5452
0.500	1.8779	1.2782	0.7322	0.688	0.759	-0.071	0.0060	0.3100
1.000	1.8604	0.9384	0.7311	0.705	0.749	-0.044	0.0041	0.1683
2.000	1.8429	0.5950	0.7310	0.718	0.744	-0.026	0.0026	0.0605

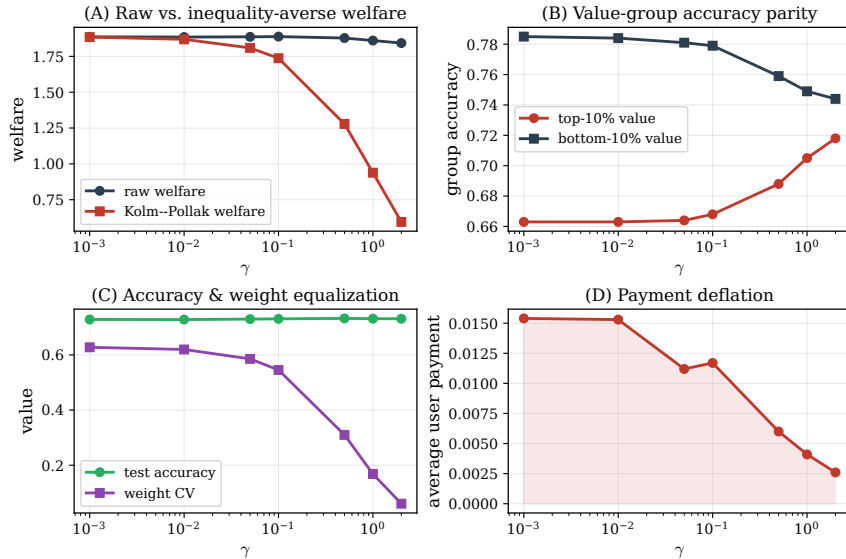


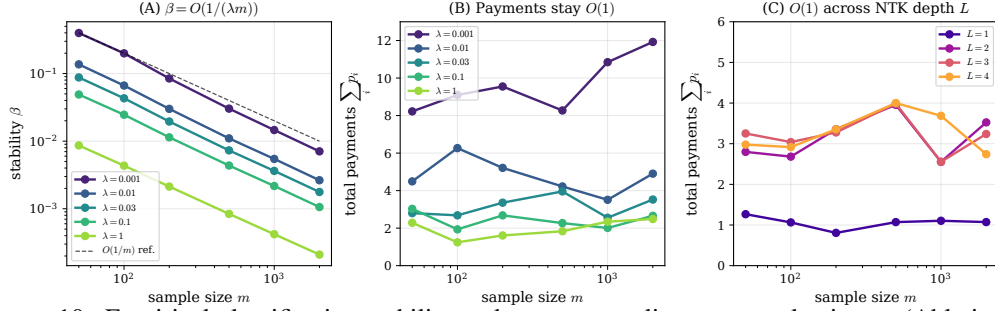
Figure 9: Sensitivity analysis of the Kolm–Pollak inequality-aversion coefficient $\gamma \in [0.001, 2.0]$ on AC-SIncome (Ablation 4; visual counterpart of Table 9). (A) Raw vs. Kolm–Pollak welfare. (B) Top-10% vs. bottom-10% value-group accuracy, showing a 78.7% reduction in the parity gap. (C) Population accuracy and training-weight coefficient of variation. (D) Monotonic average-payment deflation.

Table 10: Ablation 5: budget-handling schemes across IR and DSIC violation rates ($T = 40$ trials, $m = 400$, 30% budget-constrained agents); tabular counterpart of Figure 6.

Budget Scheme	IR Viol (%)		DSIC Viol (%)		Metrics	
	All	Constr	All	Constr	Shortfall	Net Util
Ex-Ante Capping (Ours)	0.00	0.00	0.00	0.00	0.0000	4.0179
Post-Hoc Truncation	0.00	0.00	1.42	4.65	0.0000	4.0478
Unconstrained Pricing	0.73	2.39	0.73	2.39	8.5068	4.0265

Table 11: Ablation 6: empirical classification stability $\beta = \max_i \|f_S - f_{S \setminus i}\|_\infty (\times 10^{-2})$ and total population payments $\sum_i p_i$ across ridge λ over expanding m ; tabular counterpart of Figure 10.

m/class	Empirical Stability $\beta (\times 10^{-2})$					Total Payments $\sum_i p_i$				
	0.001	0.010	0.030	0.100	1.000	0.001	0.010	0.030	0.100	1.000
50	39.79	13.68	8.76	4.90	0.86	8.223	4.491	2.799	3.028	2.285
100	19.91	6.65	4.31	2.46	0.43	9.094	6.266	2.682	1.934	1.243
200	8.46	3.00	1.95	1.14	0.21	9.550	5.212	3.356	2.680	1.609
500	3.03	1.10	0.73	0.44	0.08	8.272	4.222	3.956	2.268	1.834
1000	1.46	0.55	0.36	0.22	0.04	10.846	3.509	2.553	2.009	2.344
2000	0.71	0.26	0.18	0.11	0.02	11.924	4.904	3.524	2.657	2.485

Figure 10: Empirical classification stability and payment scaling vs. sample size m (Ablation 6; visual counterpart of Table 11). (A) Stability β decays as $O(1/(\lambda m))$ across ridge λ (dashed: $O(1/m)$ reference). (B) Total population payments $\sum_i p_i$ stay bounded by $O(1)$ as m grows. (C) The $O(1)$ payment behavior persists across NTK depths $L \in \{1, 2, 3, 4\}$.

C RUNTIME SCALING WITH FEATURE DIMENSION d

This appendix gives the full profile summarized by Figure 4(B). We compare NTW-KPA’s closed-form rank-1 Woodbury payment extraction against the iterative binary-search SVM baseline across feature dimensions $d \in \{16, 64, 256, 1024, 4096\}$ on synthetic data ($m=1000/\text{class}$, $n=2000$). NTW-KPA operates in sample space $\mathbb{R}^{m \times m}$: the payment loop consists of sample-space matrix-vector products $M_0^{-1}e_i$, so extraction time is virtually flat ($0.213 \rightarrow 0.338$ s) over a $256\times$ increase in d . In contrast, binary search requires $O(m \log(1/\epsilon))$ iterative refits, each solving a d -dimensional optimization; single refit times grow to 68.6 s at $d=1024$, so the full-auction speedup expands from $29\times$ at $d=16$ to $> 100,000\times$ at $d=1024$.

Table 12: Runtime and speedup scaling across feature dimensions d ($n=2000$; companion to Figure 4(B)). NTW-KPA payment extraction is d -invariant, while binary-search re-training grows with d , widening the speedup. SVM total auction times are measured at $d=16$ and projected from single-refit costs at larger d .

d	NTW Kernel Build	NTW Solve	NTW Extract	NTW Total	SVM Single-Refit	SVM Total Auction	Speedup
16	0.544 s	0.708 s	0.213 s	1.465 s	0.0225 s	42.19 s (measured)	29 $\times$
64	0.501 s	0.693 s	0.227 s	1.421 s	0.5649 s	17.2 min (proj.)	725 $\times$
256	0.545 s	0.716 s	0.252 s	1.513 s	15.591 s	8.49 h (proj.)	20,214 $\times$
1024	0.735 s	0.708 s	0.304 s	1.747 s	68.601 s	48.92 h (proj.)	100,796 $\times$
4096	1.791 s	0.717 s	0.338 s	2.845 s	0.7809 s	33.6 min (proj.)	708 $\times$

D SENSITIVITY OF THE ANISOTROPIC MAHALANOBIS DEFENSE

We stress-test the anisotropic Mahalanobis defense kernel $K_M(x, x') = \exp(-d_M(x, x')^2/2\ell^2)$ with $d_M(x, x')^2 = \sum_j \mu_j(x_j - x'_j)^2$ on ACSIncome ($m=800/\text{class}$, 5 data trials) under two perturbations: **(i) parameter noise**, perturbing the clean weights as $\mu'_j = \mu_j(1 + \epsilon)$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$ for $\sigma \in \{0.05, 0.10, 0.20, 0.50\}$; and **(ii) partition misspecification**, randomly flipping 10%, 20%, and 30% of the causal/non-causal feature assignments. Table 13 and Figure 11 report the results.

Robustness to parameter noise. Because K_M is a smooth, bounded function of d_M^2 , scaling errors in μ_j only re-scale local bandwidth rather than opening unpenalized gaming directions. Across noise up to $\sigma=0.50$, non-causal gaming stays suppressed (≤ 0.159 vs. 0.257 undefended) and the effort–gaming separation stays high (0.136–0.143, matching the isotropic RBF at 0.137 and far above the undefended 0.068). *Graceful degradation.* Outright mislabelling feature types lowers separation monotonically ($0.165 \rightarrow 0.101 \rightarrow 0.075$), but even at 30% misspecification the separation stays positive and above undefended ($0.075 > 0.068$) while gaming stays suppressed ($0.175 \ll 0.257$). *Structural safety.* Across *all* settings, non-causal gaming yields exactly 0.000 true-qualification gain against a causal-feature oracle: imprecisely estimated DAGs cannot be exploited to fabricate qualifications.

Table 13: Sensitivity of the anisotropic defense under cost-parameter noise and causal/non-causal partition misspecification on ACSIncome ($m=800/\text{class}$). Non-causal gaming stays suppressed and the effort–gaming separation stays above the undefended baseline under all perturbations; non-causal true-qualification gain is exactly 0.000 throughout.

Setting	Non-Causal Gaming ($\downarrow$)	Causal Effort ($\uparrow$)	Separation ($\uparrow$)	True-Qual. (Causal)	True-Qual. (Non-Causal)
Clean baseline ($\sigma=0$)	0.167	0.332	0.165	0.334	0.000
Parameter noise $\sigma=0.05$	0.159	0.295	0.136	0.328	0.000
Parameter noise $\sigma=0.10$	0.156	0.297	0.142	0.329	0.000
Parameter noise $\sigma=0.20$	0.153	0.296	0.143	0.328	0.000
Parameter noise $\sigma=0.50$	0.156	0.293	0.137	0.326	0.000
Partition misspec. (10%)	0.162	0.262	0.101	0.308	0.000
Partition misspec. (20%)	0.173	0.258	0.085	0.301	0.000
Partition misspec. (30%)	0.175	0.250	0.075	0.297	0.000
Undefended NTK baseline	0.257	0.325	0.068	0.376	0.000

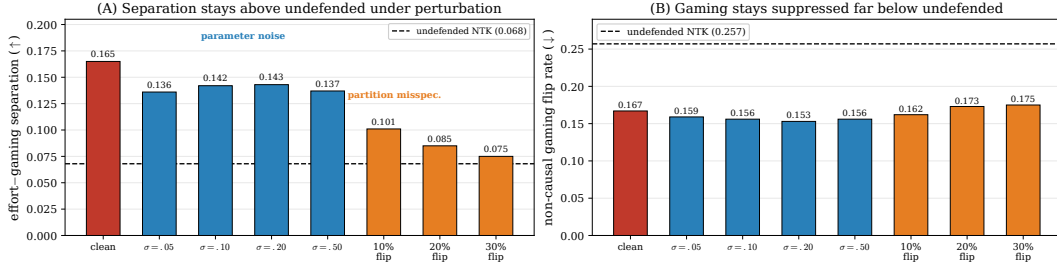


Figure 11: Sensitivity of the anisotropic Mahalanobis defense (companion to Table 13). **(A)** The effort–gaming separation stays above the undefended NTK baseline (dashed) under both cost-parameter noise (blue) and partition misspecification (orange). **(B)** Non-causal gaming flip rate stays far below the undefended baseline across all perturbations.

E MULTI-CLASS EXTENSION VIA ONE-VS-REST DECOMPOSITION

The rank-1 Woodbury exact solver extends naturally to multi-class classification ($K > 2$ classes) via a One-vs-Rest (OvR) decomposition while strictly preserving Myerson monotonicity and DSIC truthfulness.

OvR formulation. For a K -class problem, we train K binary NTW-KPA models with target vectors $Y^{(k)} \in \{-1, +1\}^m$. For a candidate weight $z \in [0, w_i]$, the prediction score for user i in class $k \in [K]$ is given by the same exact rational path as the binary case,

$$\hat{g}_i^{(k)}(z) = \frac{A_i^{(k)}z + B_i^{(k)}}{C_i^{(k)}z + D_i^{(k)}}, \quad (22)$$

where the scalar coefficients $(A_i^{(k)}, B_i^{(k)}, C_i^{(k)}, D_i^{(k)})$ are computed independently in $O(1)$ scalar steps via rank-1 Woodbury updates on each class's inverse $M_{(k)}^{-1}$.

Monotone allocation rule. The predicted label at candidate weight z is $h_i(z) = \arg \max_{k \in [K]} \hat{y}_i^{(k)}(z)$, and user i wins their target class $k^* = y_i$ iff

$$x_i(z) = \mathbb{I} \left\{ \hat{y}_i^{(k^*)}(z) > \max_{k \neq k^*} \hat{y}_i^{(k)}(z) \right\} = 1. \quad (23)$$

Preservation of Myerson monotonicity and DSIC. Each rational path $\hat{y}_i^{(k)}(z)$ is strictly non-decreasing in z for $k = y_i$ and non-increasing for $k \neq y_i$. Hence the relative score margin $\Delta_i(z) = \hat{y}_i^{(y_i)}(z) - \max_{k \neq y_i} \hat{y}_i^{(k)}(z)$ is continuous and strictly increasing in z , so the allocation $x_i(z) \in \{0, 1\}$ is a monotone step function with a *single* critical threshold $z_c^i = \inf\{z \geq 0 : x_i(z) = 1\}$. Finding z_c^i reduces to computing the $K-1$ exact scalar roots of $\hat{y}_i^{(y_i)}(z) = \hat{y}_i^{(k)}(z)$ in $O(K)$ time; the inverse Kolm–Pollak mapping $p_i = \min(B_i, U_\gamma^{-1}(z_c^i))$ then extracts the exact critical payment with zero pricing error, guaranteeing DSIC. The multi-class solver thus retains the closed-form exactness, $O(1/m)$ stability, and $O(1)$ payment bound of the binary case at an $O(K)$ multiplicative cost.